Initial Project Report

Handwritten Receipt Recognition and Information Extraction System

1. Understanding of the Problem Statement

The objective of this project is to develop a system that can read handwritten receipts, recognize the text, and convert the information into a structured digital format.

Handwritten receipts can be difficult to process because of unclear handwriting, poor image quality, faded ink, crumpled paper, and inconsistent formats. Manually entering receipt details into a database is time-consuming and may lead to errors.

The proposed system aims to automate this process using Optical Character Recognition (OCR), Handwritten Text Recognition (HTR), Natural Language Processing (NLP), and machine learning models.

The system will identify important details such as store name, date, purchased items, individual prices, and total amount. The extracted information can then be stored in a database, making it easier to search, retrieve, and analyze receipt data.

2. Proposed Project Approach

The project will be developed in the following phases:

Phase	Description
Phase 1: Data Preparation	Collect suitable handwritten receipt datasets, clean the images, and prepare annotated data for model training and testing.
Phase 2: Text Detection	Explore and implement models such as DBNet or CRAFT to identify text regions in receipt images.
Phase 3: Handwritten Text Recognition (HTR)	Explore pretrained models such as TrOCR or CRNN to recognize handwritten text and convert it into digital text.
Phase 4: Information Extraction (KIE)	Use NLP or layout-aware models such as LayoutLMv3 to extract structured details, including store name, date, items, prices, and total amount.
Phase 5: Evaluation and Deployment	Evaluate the system using Character Error Rate (CER) and F1-score, store the extracted data in a database, and develop a basic interface or API for testing.

3. Proposed Timeline (2 Months)

The following is a tentative 8-week timeline. The schedule may be adjusted based on dataset availability, model performance, and project requirements.

Duration	Planned Activities	Expected Outcome
Week 1–2	Understand the problem statement, research existing techniques, explore Hugging Face models, and identify suitable datasets.	Initial research, dataset shortlist, and model selection.
Week 3–4	Prepare the dataset, perform image preprocessing, and implement a baseline text detection and OCR pipeline.	Prepared dataset and initial text recognition results.
Week 5–6	Explore and implement HTR and information extraction techniques to identify important receipt fields.	Extracted receipt details in a structured format.
Week 7	Integrate the pipeline with a database and develop a basic interface or API for uploading and processing receipts.	Working prototype with data storage and retrieval.
Week 8	Evaluate model performance, perform testing, improve results where possible, and prepare project documentation and final demonstration.	Evaluated prototype and final project report.

4. Expected Project Outcome

At the end of the internship, the expected outcome is a working prototype that can process handwritten receipt images, recognize text, extract important information, and store the results in a structured database. The system will aim to reduce manual data entry and make receipt information easier to access and analyze. The final performance will depend on the quality of the dataset and the accuracy of the selected models.

5. Initial Learning and Research Requirements

To successfully complete the project, the following areas will be explored during the internship:

- Python and essential machine learning libraries
- Computer Vision and image preprocessing using OpenCV
- OCR and Handwritten Text Recognition (HTR)
- NLP and Information Extraction (KIE)
- Pretrained and open-source models available on Hugging Face
- SQL and database management
- LLMs, embeddings, and vector databases for querying receipt information, if required
- Model evaluation and basic API/UI development

6. Conclusion :- The project will begin with research, dataset identification, and exploration of existing pretrained models. Based on the initial findings, a suitable approach will be selected and implemented step by step. The primary focus will be on developing a functional prototype within the two-month internship period, with regular progress updates and improvements based on the results obtained during each phase.